HKU DBF TEAM EVENT RECORD 2024 – 2026
&
HKU AIAA Student Branch Application Proposal



Sponsors:

Tam Wing Fan Innovation Wing
FACULTY OF ENGINEERING

Ansys

Department of Mechanical Engineering
The University of Hong Kong

L'VOYAGE

The Safety Collaborative

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Executive Summary

The team leader of HKUDBF 2026 (Zhao Zehui), on behalf of the whole team, would like to introduce the events organised by HKUDBF from year 2024-2025 and the team's competition history. Each year, the final-year students in HKUDBF would participate in AIAA DBF competition as the topic of the final year project. In AIAA DBF 2024 – 2025, the team, formed by 5 year-4 students, received 40th out of 112 teams, and in AIAA DBF 2025 – 2026, the team received 32nd out of 98 teams. The team would like to request for establishing AIAA student branch from 2026 – 2027, to organize AIAA related events to provide more information and relationships between the team members and professionals in AIAA. AIAA, full name as American Institute of Aerospace and Aviation, offers lectures, online courses and competitions for students worldwide every year to enhance their understandings, passions and skills in aerospace and aviation industry. The AIAA Student Branch will be formed by student members from Year 3 to 4 in year 2026 to 2027.

In AIAA DBF 2025 – 2026, the team secured 2 sponsorships from Innovation Wing, Department of Mechanical Engineering, Faculty of Engineering, The University of Hong Kong; 2 sponsorships from aviation companies, called L'Voyage, a private jet company, and the Safety Collaborative, a human factor consultancy; and lastly a license sponsorship from ANSYS. The total sponsorship amount exceeds 109,000 HKD, excluding the ANSYS license. The team are also participating in 2 CUADC competitions in year 2026-2027. CUADC, full name as China Universities Aircraft Design Competition, which is related to STOL (Short-take-off-landing) and solar powered airplane design, build and fly. Both teams are formed by 4 Year-1 and Year-2 students.

Lastly, the team is also going to participate in another competition called AIAA Air Strike Carrer-based Aircraft Competition 2026, which is a competition that focused on proposal and report. The competition team is formed by 2 year-3 students, 1 year-2 students, and 2 year-1 students. HKUDBF 2026 aims to provide more opportunities for students from all year to participate in competition projects to gain experiences and skills in aviation field.

Introduction

The HKU Design, Build & Fly (DBF) Team, comprised of undergraduate students from the University of Hong Kong supervised by Dr CK Chan, is founded in 2010 and collaborated with Boeing Company to design a model aircraft to fly in the air for 6 hours. Since then, the team participated in the national and international model aircraft competitions every year, and

organized student-initiated courses and hands-on workshops to provide aircraft-related knowledge and practical applications into aircraft design and manufactures.

This document includes the event record of HKUDBF 2024 – 2026, and proposed Events for HKU AIAA student branch from 2026 – 2027.

Projects from 2024 - 2026

Aircraft Design for AIAA DBF Competition 2025 (FYP Student Team)

The DBF Team – AIAA has developed a 7kg-model aircraft for the AIAA 2025 competition this year which is capable of a minimum operation duration of 6 minutes. This model aircraft is subjected to balance performance and manufacturability. Address and optimize payload complexity, develop an efficient fuel tank management system, and achieve bonus objectives related to X-1 test vehicles. The model aircraft consists of several components, including Wing, Propulsion system, Empennage, Pylon, Landing Gear, Fuselage and X-1 Glider, each comprised of self-developed technical features.

i. Pylon: two-piece of external 3D-printed pylon and internal mounting constructed underneath the wing, which is secured by 3M VHB tape, with pylon, screws and carbon-fiber rod and wing spar
ii. X-1 Glider: It is a tailor-made 3D printed glider being released by the main plane during a mission this year, which is required to land in a specific spot. A waypoint navigation system (a combination of GPS, gyroscope and an anemometer) and an easy-release mechanism are created by us.
iii. Avionics and Control system: Our transmitter/ receiver contains 7 channels (2× aileron, elevator, rudder, propulsion, release, UBEC)

Impact of these technologies

To minimize the weight of the X-1 glider, different foaming 3D printing filaments were tested this year. Foaming ASA and PLA were utilized as it may perform almost half weight reduction at the same volume of the printing model. This characteristic can helps to perform better in mother-aircraft's flying performance by reducing its payload. As these filaments are far more difficult to handle compared to ordinary non-foaming filaments, we performed many material testing and printing during manufacturing. As a result, our team has accumulated abundant 3D printing experience. Furthermore, the X-1 plane we developed could be used in various applications, for example, quick delivery in low-altitude economies. Our material research and testing offer detailed references for all members.

Recognitions or awards received

The HKU DBF Team ranked 40 out of 112 university teams from around the world in 2024 – 2025.



AIAA DBF Competition 2026

In AIAA DBF 2026, we are going to design an aircraft with 1.25-m wingspan aircraft carries a payload consisting of 3 passengers and 1 cargo and a 4-m banner to complete three flight missions and one ground mission. Mission 1 (M1) is a delivery flight that demonstrates the aircraft's flight capabilities without a payload. Mission 2 (M2) is a carriage flight that includes passengers and cargo. Mission 3 (M3) is a launch flight that demonstrates the stowing, towing, deployment and release of banner. We designed and manufactured 3 prototypes and performed 10 flight tests, each with their own unique features to achieve the best possible score we can have in the competition.



Prototype 1 (January 2026)



Prototype 2 (February 2026)



Prototype 3 (March 2026)



Banner Towing Event (March 2026)

Aircraft Configuration:

i. Wing: Fuel Tank Reinforcement by using small rips and spars with plywood and balsa wood material to achieve a lightweight wing design and withstand external fuel tank weight of 1.46kg designed by 3D printing technique. Several 3D printed parts are designed and implemented to improve wing performance.
ii. Propulsion system: Self-designed thrust testing to maximize the thrust-to-weight ratio
iii. Empennage: elevator and rudder sized for low-speed maneuverability, with self0-designed flight test to test and improve aircraft turning by enlarging rudder surface
iv. Landing Gear: Aluminum taildragger landing gear with sufficient clearance and impact-taking ability is designed to perform static loading test, drop test, displacement simulation and stress simulation.

v. Fuselage: formers, firewall, X-1 and MLG attachment plate, Longerons, truss structure, wing seat and internal tank fixture are used to construct our fuselage, which transfers loading/ balance the plane's weight and ensures strong structural integrity and operational adaptability for multiple missions.
vi. Banner Deploy and Release Mechanism: Make use of gripper mechanism design to stow the folded banner underneath the aircraft, deploy and tow the banner using remote-control, and release the banner during flight using 30-35-kg torque servo.

AIAA Aircraft Design Competition

Other than the AIAA DBF competition for final year students to participate, we also encouraged non-final year members to join other competitions so they can gain experiences and be exposed to more opportunities. The AIAA Aircraft Design competition 2026 is a written competition which requires teams to write a proposal and a report for preparing an air strike carrier-based aircraft. The team is led by year-3 students, with 6 team members.

CUADC STOL Competition 2026

This is a team formed by 4 year-1 team members to participate in a Short-take-off and landing competition held by CUADC in China. The team members have been working on aircraft design and performing flight test since February 2026.



CUADC Solar Plane Competition 2026

There are 2 year 2 members and 3 year 1 members in participating this competition. The team has to installed at least 12 solar panels to provide power for the aircraft to manoeuvre in the air during the flight.

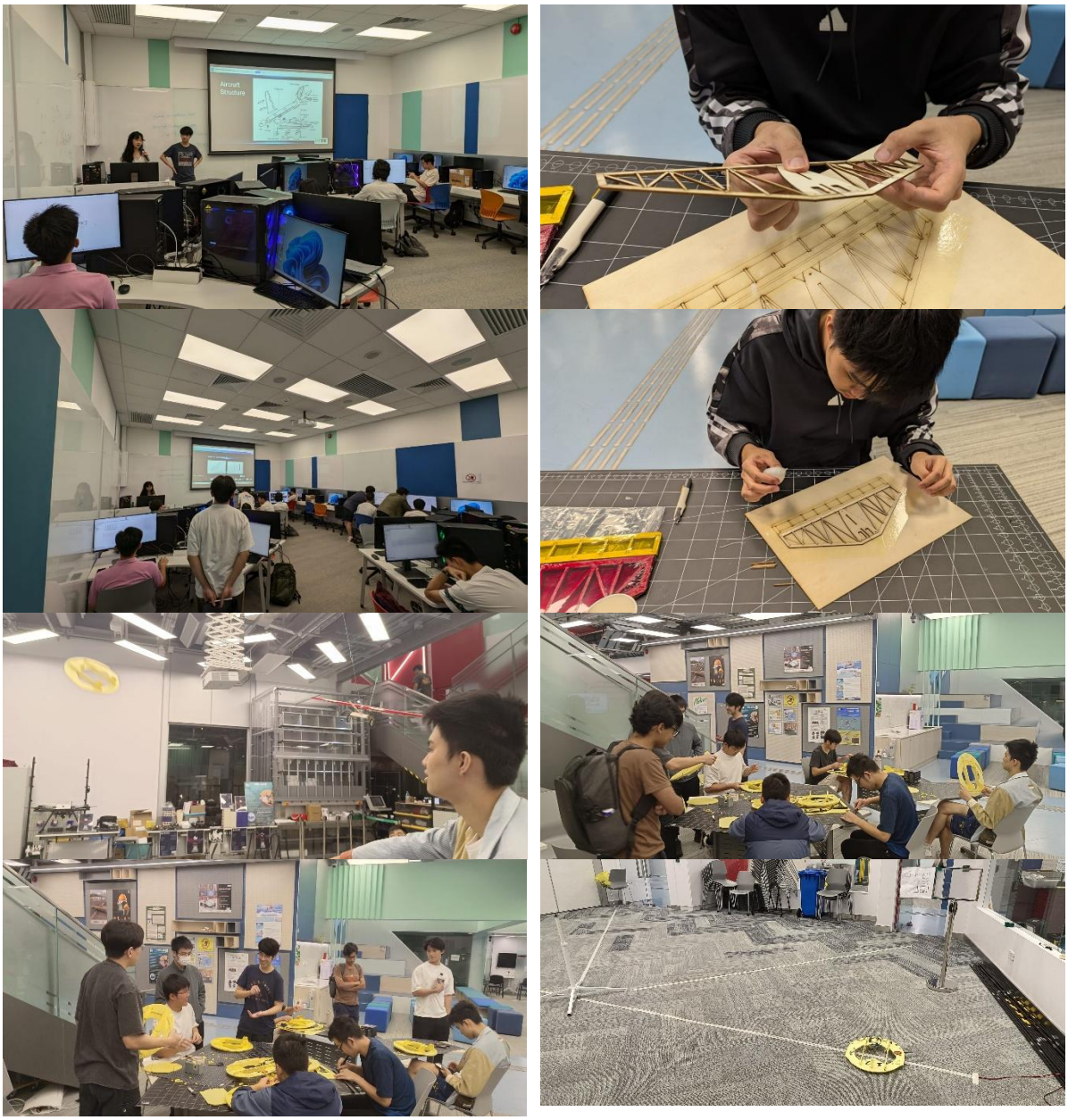
Events and Visits in 2024 - 2026

Workshops for undergraduates

In 2024-2025 years, the DBF team has organised regular workshops (students-initiated-courses) for students throughout semester 1 and 2. Students acquired technical skills from workshops,

such as SolidWorks model design, aircraft manufacturing and round-the-pole foam plane making.

Photos for workshops in semester 1, 2024-2025:



Photos for Workshops in semester 2, 2024-2025:







Aerial Master The Ultimate Flight Simulator Challenge

Other than workshops and courses, DBF team has also organised Aerial Master The Ultimate Flight Simulator Challenge, a fun contest for students to experience virtual fighter jet maneuverer and win mini aircraft model figure as winning prizes.

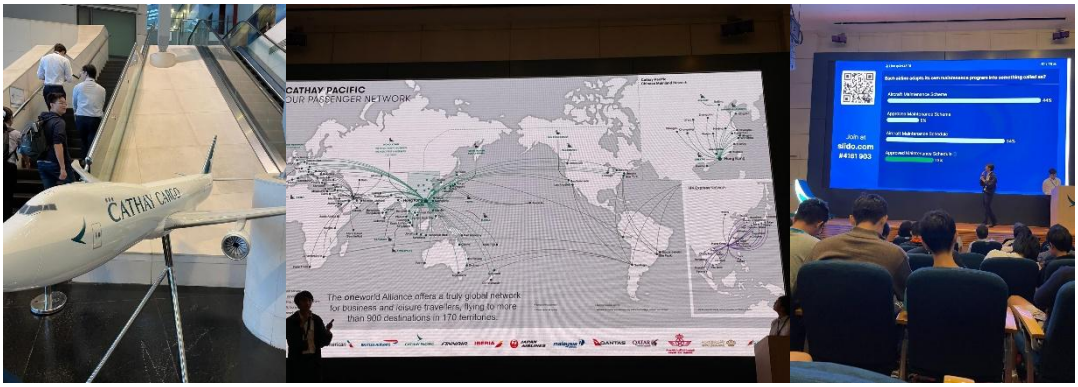


HKUDBF Recruitment talk

At the start of the semester 1 in 2024 (27 September), the team organised a recruitment talk to promote the team's exciting activities and attract new members, including the AIAA competition, BMFA competition and different hands-on skills and CAD/CAM skills that will be acquired through workshops. Until June 2025, the DBF team has recruited more than 30 new members, including undergraduates and postgraduates.

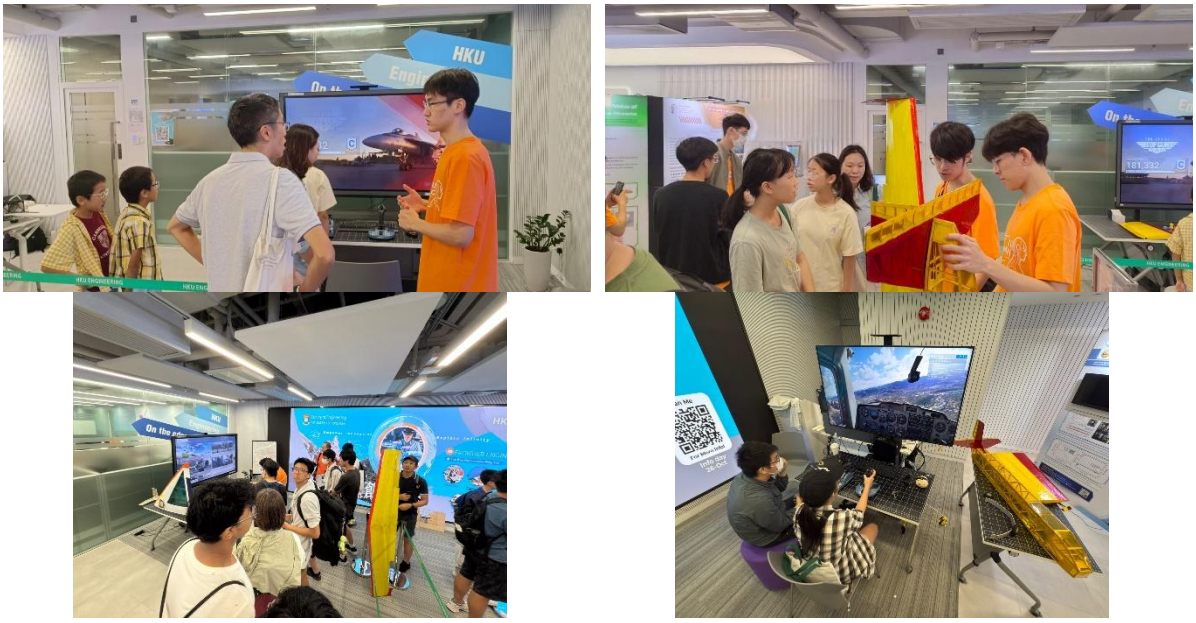
Visit to Cathay Pacific City

The team also helped promote a visit to Cathay Pacific City on 13 November 2024 (which is organised by the faculty) by organising and gathering students to attend the event. It allowed students to understand the career in Cathay Pacific by the talk offered by seniors and by chatting with them in small groups. Moreover, participants could take a glance at the places the staffs are trained, work and lived.



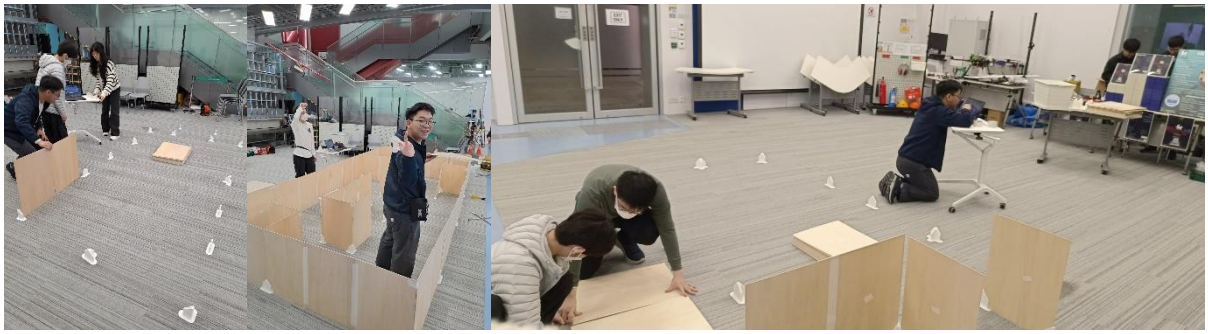
HKU Information Day

On 26 October, the DBF team held booth at Innovation Wing Two to showcase aircraft model from previous years' design and placed socket drones and two flight simulators for visitors to experience and enjoy the fun of aircraft control on HKU info day. Visitors include parents, students and teachers.



FPV Workshop in Innovation Wing

On 24 January 2025, the UAS team has collaborated with DBF team to construct a maze for FPV drone and welcome all innovation wing members and non-innovation wing members to experience FPV actual and virtual drones' control.



Alumni Day

On 16 March 2025, the DBF team promoted DBF by explaining how DBF are being an innovative team to both Alumni and journalist. Socket drones and flight simulators (RC and commercial plane) were set at DBF team's booth for visitors to play.



DBF Activity 2025 - 2026

In year 2025-2026, the DBF AIAA team has been performing preparation including meeting, manufacturing, testing, new product design based on next year's competition requirements. The DBF team also continues organising workshops and trainings for undergraduates, knowledge exchange with undergraduates and postgraduates, and visits to professional institutions and companies.

Summer Workshops and Recruitment Talks (Jun to Aug 2025)

We invited our senior who participated AIAA in 2024 – 2025 to give us a sharing on their competition experiences.



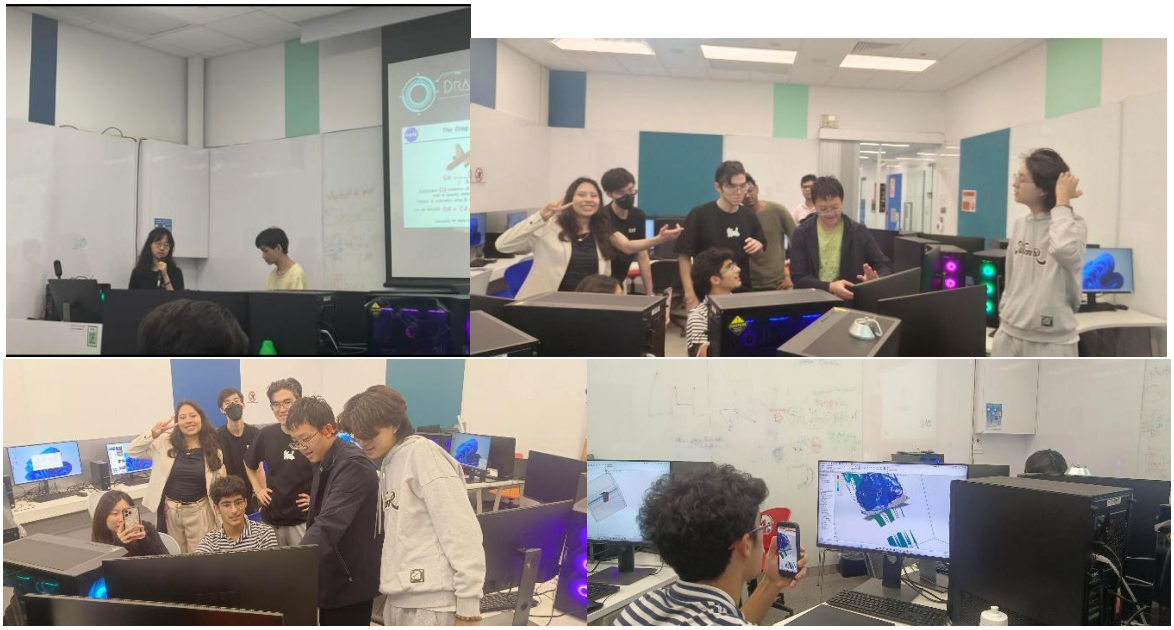
We introduced our team to freshmen and interested students.





Semester-Long Workshop Series (Sept to Nov 2025)

Our semester workshops introduce basic aerodynamic knowledge, aircraft design and CAD modelling on SolidWorks, and CFD analysis on SolidWorks.



Members who actively participated in our workshops will be selected for a interview to join other competitions and be the helpers of AIAA DBF team.



HKU Info Day (Oct 2025)

The DBF team held their booth in university to promote aviation to the public and encourage students to learn more about team's activities and university life in HKU.


Please Queue Up
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HKUDBF
WE DESIGN, BUILD AND FLY
TEAM OF AIAA & BMFA
2025
AIAA
START
ADVENTURE

Manufacturing Training Sessions

As we started manufacture our first aircraft in January 2026, we also recruit helpers who actively participated in HKUDBF workshops in semester 1, 2026. 9 helpers from Year 1 to 3 assisted us in manufacturing all the 3 aircrafts.







Our previous year's events: DBF Team in 2023 – 2024



Sponsorships and Financial Budget

In AIAA DBF 2026, we secured 5 sponsorships, with Tier level from Bronze, Silver, Gold and Diamond, and Engineering:

Sponsors	Tier Level	Amount of Cost
Innovation Wing, Faculty of Engineering, The University of Hong Kong	Diamond	HK$ 100,000

Ansys	Engineering	A full set of License (workbench, mechanical, CFD etc)
Mechanical Engineering Department, Faculty of Engineering, The University of Hong Kong	Diamond	HK$ 6,000 Plus Accommodation Fee and Transportation (HK$ 30,000)
L'Voyage	Silver	HK$ 3,000
The Safety Collaborate	Bronze	HK$ 1,000
Total		HK$139,000 +

Upcoming Schedule and Events

April	• Design/Build/Fly Competition Flyoff in Wichita, Kansas, USA • Setup AIAA Student Branch
June-August	• AIAA Aircraft Design Competition • CUADC STOL & Solar Panel aircraft Competition • Training workshops for next year AIAA members • Workshops and educational activities for secondary school students

Chartering AIAA Student Branch

The full name of AIAA is American Institute of Aeronautics and Aerospace. The HKU AIAA Student Branch would provide a platform for students to share their passion in aviation and aerospace in the form of knowledge exchange, competition, papers and presentation, networking and workshops, and accessing educational resources and advice from AIAA professionals on AIAA website.

Benefits of chartering AIAA Student Branch

By chartering AIAA student branch, members would have to opportunities to access several Official recognitions as an AIAA-affiliated organization:

- Access to AIAA scholarships, competitions, and technical programs
- Eligibility to participate in Regional and International Student Conferences
- Access to the Distinguished Speaker Series
- Connection to AIAA professional sections and global aerospace networks
- Leadership development opportunities for student officers

Proposed Events for AIAA Student Branch

Some of the most significant events will be the main focus, including AIAA Distinguished Speaker Series, where we can invite professionals in Aviation and aerospace industries to give a virtual talk to the members, resume building and professional development suggestion from

professionals in AIAA, and many more events for members to build up one another's relationship and bonding. The proposed events for AIAA student branch 2026 – 2027 includes:

1. Socials, trivia or movie night
2. Field trips
3. Guest speaker or panel
4. Project showcase or student paper presentations
5. Resume building and professional development
6. Mentoring/networking
7. Model building
8. Lunch and learn
9. Aircraft design workshops and manufacturing training sessions
10. Watch parties for online events
11. AIAA Distinguished Speaker Series - AIAA Distinguished Speaker Series | AIAA

Proposed Name list of the Committee of HKUDBF AIAA Student Branch

The following table includes the proposed roles and student's name who are going to be the committee of HKUDBF AIAA Student Branch 2026 – 2027.

ROLES	AIAA STUDENT BRANCH	Faculty Advisor	Founding Advisor
Chair	LAM Tsun Hei Howard	Dr Chan	Zhao Zehui, Leila
Vice-chair	Adriana Cecilia Zurita Pimentel		
Secretary	Poon Chapman		
Treasurer	Muhammad Nusayr Bin Rahman		
Membership Officer	Lei Ka Cheng		
Media officer	Adriana Cecilia Zurita Pimentel		

The founding advisor is the team leader of HKUDBF team in 2025 – 2026. The role and responsibility of the founding advisor include:

- providing my competition experiences and event holding experiences,
- providing the documents and information I collected for the team,
- connections I have made with other universities, and seniors of our team.

The team includes team members from year 1 to 4 so that each year, there will be new members joining, and the old members with sufficient experiences will continuously run the team.